Notes: Collard-Wexler & De Loecker (AER, 2015)

Reallocation and Technology: Evidence from the US Steel Industry

Contents

1 Big picture

- **Question.** How much of the dramatic productivity growth in US steel came from a new production technology — the minimill — and how much came from reallocation across producers induced by that technology?
- **Setting.** US steel plants, 1963–2002. The industry is ideal because output is remarkably stable in product composition over time, so productivity growth is mainly about *process innovation* rather than new goods.
- **Core insight.** The arrival and diffusion of minimills changed the competitive environment of the industry. That change increased aggregate productivity through two channels: direct displacement of vertically integrated producers and induced reallocation among the surviving vertically integrated plants.
- **Main result.** Aggregate steel productivity rose sharply while employment collapsed. In the authors' plant-level decomposition, aggregate TFP rises by about 22% from 1963 to 2002, and roughly one-third of that increase comes from reallocation toward more productive producers. A large share of this reallocation is specifically between technologies, from VI plants toward MM plants.
- **Technology channel.** Minimills are about 7.5% more productive than vertically integrated plants in the preferred specification. Reallocation from VI to MM explains about 23% of industry productivity growth, and improvements within minimills account for roughly another fifth.
- **Competition channel.** The old technology does not simply disappear. Instead, intensified competition from minimills pushes out less efficient VI plants, especially in bar products, so surviving VI producers become much more productive through within-technology reallocation.
- **Broader implication.** A new technology can raise aggregate productivity not only because it is intrinsically better, but also because it changes product-market competition and the allocation of output across plants.

2 Data and measurement (key objects)

2.1 Plant-level steel data

- **Sources.** Census of Manufacturers (CMF), Annual Survey of Manufacturers (ASM), Longitudinal Business Database (LBD), plus product and material trailers.
- **Coverage.** US plants producing carbon or alloy steel, 1963–2002.
- **Unit of observation.** Plant-year, with product-level revenues and quantities for output, and plant-level inputs, employment, capital, and materials.
- **Why this industry works well.** Steel products in the 1960s and 2000s are very similar, so growth is not mostly a "new goods" story. This sharply simplifies interpretation of productivity growth.

2.2 Technology classification

- **Vertically integrated plants** (VI)**.** Steel is produced through blast furnaces and basic oxygen furnaces, using iron ore, coke, and limestone to make pig iron and then steel.
- **Minimills** (MM)**.** Electric arc furnaces melt scrap steel and direct reduced iron. These plants are much smaller in efficient scale and historically specialized in lower-quality products, especially bars.
- **Plant technology is effectively fixed.** A plant does not switch from VI to MM, which is useful for identification.
- **Classification method.** For 1992, 1997, and 2002 the authors use a direct steel questionnaire; for earlier years they infer technology from materials and products.

2.3 Products, prices, and quality adjustment

- **Product categories.** The paper works with detailed steel product groups such as hot rolled sheet, cold rolled sheet, hot rolled bars, steel ingots and semifinished products, pipe and tube, wire, cold-finished bars, and blast-furnace related products.
- **Output prices.** The authors construct plant-specific output price deflators from product-level BLS prices matched to each plant's output mix.
- **Input prices.** They also construct plant-specific materials price indexes, important because minimills and integrated plants buy very different intermediate inputs.
- **Key measurement lesson.** Controlling for plant-level output prices is crucial. Without it, productivity comparisons between MM and VI plants are badly biased because integrated plants are more active in higher-price sheet products.

2.4 Main descriptive facts

- **Steel versus manufacturing (Table 1).** From 1972 to 2002, steel-sector TFP rises by 28%, while shipments fall by 35%, labor falls by 80%, and output prices fall by 23%.
- **Industry evolution (Figure 1).** Shipments recover after the early-1980s collapse, but employment keeps falling. Total assets decline, while the number of minimills rises and the number of integrated plants falls.
- **Changing product space (Figure 2).** Minimills capture bars and ingots over time, but only enter sheet products in a meaningful way much later.
- **Price patterns (Figure 3).** Output prices of steel products fall sharply after the early 1980s. Input prices spike in the late 1970s and early 1980s, with bigger increases for integrated plants because they are more energy intensive.
- **Entry and exit (Table 2).** The 1980s show simultaneous large MM entry and VI exit, suggesting an important role for technology-driven reallocation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Technology-specific production function

For a plant i, product j, and year t, output under technology $\psi \in \{\mathrm{VI}, \mathrm{MM}\}$ is

$$Q_{ijt} = F_{\psi,t}(L_{ijt}, M_{ijt}, K_{ijt}) \exp(\omega_{it}). \tag{1}$$

- **Interpretation.** The two technologies can have different production structures, but productivity ω_{it} is plant-specific and Hicks-neutral.
- **Economic point.** This formulation lets the authors test whether minimills are better because of higher plant productivity, different production elasticities, or both.

3.2 Empirical Cobb-Douglas representation

The product-line production function is specialized to Cobb-Douglas:

$$Q_{ijt} = X_{ijt}\,\Omega_{it}, \tag{2}$$

where $X_{ijt} = L_{ijt}^{\beta_l} K_{ijt}^{\beta_k} M_{ijt}^{\beta_m}$. Because inputs are observed only at the plant level, product-line inputs are allocated using product revenue shares. After deflating output and materials, the plant-level estimating equation becomes

$$\tilde{q}_{it} = \beta_l l_{it} + \beta_m m_{it} + \beta_k k_{it} + \omega_{it} + \varepsilon_{it}. \tag{3}$$

- **Interpretation.** $\tilde{q}_{it}$ is plant-level output net of plant-specific output prices; m_{it} is materials net of plant-specific input prices.
- **Key coefficients in the preferred specification.** Roughly, materials receive elasticity 0.68, labor 0.274, and capital 0.079.
- **Why these deflators matter.** Omitting output-price corrections compresses the measured MM productivity premium because VI plants disproportionately produce high-price sheet products.

3.3 Investment control and selection correction

Following Olley and Pakes (1996), investment is used to proxy for unobserved productivity:

$$i_{it} = i_t(k_{it}, \omega_{it}, \psi_i), \tag{4}$$

which is inverted to write $\omega_{it} = h_{\psi,t}(k_{it}, i_{it})$. The first-stage control-function equation is

$$\tilde{q}_{it} = \phi_{\psi,t}(l_{it}, m_{it}, k_{it}, i_{it}) + \varepsilon_{it}. \tag{5}$$

The second stage imposes a productivity law of motion and moments of the form

$$\mathbb{E}\left[\xi_{it}(\beta)\begin{pmatrix} l_{i,t-1} \\ i_{i,t-1} \\ k_{it} \end{pmatrix}\right] = 0. \tag{6}$$

- **Interpretation.** This addresses simultaneity (inputs responding to productivity) and selection (low-productivity plants exiting).
- **Plant state.** The inclusion of ψ_i in the state vector allows the dynamic process to differ across technologies.

3.4 Static productivity decompositions

Aggregate productivity is decomposed using Olley-Pakes:

$$\Omega_t = \bar{\omega}_t + \Gamma_t^{\mathrm{OP}}, \tag{7}$$

where Γ_t^{OP} is the covariance between market share and productivity. The authors then separate this into within-technology and between-technology pieces. The within decomposition is

$$\Omega_t = \sum_{\psi \in \{\mathrm{MM}, \mathrm{VI}\}} s_t(\psi)\big(\bar{\omega}_t(\psi) + \Gamma_t^{\mathrm{OP}}(\psi)\big), \tag{8}$$

and the between-technology decomposition is

$$\Omega_t = \bar{\Omega}_t + \Gamma_t^{B}, \tag{9}$$

where Γ_t^B captures reallocation across minimills and integrated plants.

- **Interpretation.** Γ_t^{OP} measures whether bigger producers are more productive; Γ_t^B measures whether the industry shifts weight toward the better technology.
- **Big empirical takeaway.** The paper is not just showing that minimills are more productive. It shows that output share moved toward that technology in a quantitatively important way.

3.5 Dynamic decomposition with entry and exit

To isolate the roles of incumbents, entrants, and exiters, aggregate productivity growth is written as

$$\Delta\Omega_t = \underbrace{\sum_{i\in\mathcal{A}} s_{i,t-1}\Delta\omega_{it}}_{\text{plant improvement}} + \underbrace{\sum_{i\in\mathcal{A}} \omega_{i,t-1}\Delta s_{it}}_{\text{reallocation}} + \underbrace{\sum_{i\in\mathcal{B}} s_{it}\omega_{it}}_{\text{entry}} - \underbrace{\sum_{i\in\mathcal{C}} s_{i,t-1}\omega_{i,t-1}}_{\text{exit}}. \tag{10}$$

- **Interpretation.** Productivity growth comes from improvements within incumbents, shifting market shares among incumbents, the productivity of entrants, and the removal of low-productivity exiters.

- **Why this matters.** This decomposition clarifies whether minimills improve by learning and technical progress, while integrated producers improve mainly through competitive selection.

3.6 Markup measurement

Using the De Loecker and Warzynski (2012) logic, the technology-specific markup is measured as

$$\mu_t(\psi) = \beta_m \frac{R_t(\psi)}{E_t(\psi)}, \tag{11}$$

where $R_t(\psi)$ is total revenue and $E_t(\psi)$ is total expenditure on variable inputs for technology ψ.

- **Interpretation.** A fall in $\mu_t(\psi)$ indicates stronger competition and a more elastic residual demand curve.
- **Role in the paper.** Falling markups provide direct evidence that the arrival of minimills increased product-market competition, especially in the bar segment.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- **Exogenous technological arrival.** The diffusion of minimills is treated as the arrival of a new process technology in steel. Plants do not switch from VI to MM, which simplifies the comparison.
- **Stable products, changing process.** Because steel products are similar over time, the productivity story is not mainly about changing product composition at the industry level.
- **Long horizon.** The 1963–2002 panel is long enough to capture slow entry, exit, and market-share reallocation, which are central to the mechanism.

4.2 Why the productivity estimates are credible

- **Simultaneity and selection.** The Olley-Pakes investment proxy is used to control for endogenous input choices and nonrandom exit.
- **Output prices.** Product-specific BLS prices are used to purge plant productivity estimates of product-mix and quality-price effects.
- **Input prices.** Plant-specific material price indexes account for the fact that MM and VI technologies use different energy and material bundles.
- **Technology-specific production functions.** The paper allows production elasticities to vary by technology and shows results are similar, so the main findings are not driven by forcing the same production function on both technologies.

4.3 Alternative explanations the paper addresses

- **Management and ownership.** The minimill premium survives firm-year fixed effects, so it is not simply that better firms own minimills.
- **Geography.** The premium also survives state-year and firm-state-year controls, so it is not just a regional story.
- **International competition.** Steel did not face unusually large import growth relative to manufacturing as a whole, so trade alone cannot explain the steel sector's extraordinary productivity growth.
- **Unionization.** The paper argues that unionization differences are unlikely to account for the findings. Within-firm-state-year comparisons still show a sizable minimill productivity advantage.

5 Tables and figures

5.1 Table 1: Relative performance of steel versus manufacturing

Read:

- Steel is unusual because productivity rises while the sector contracts sharply in shipments and especially in labor.
- Relative to typical manufacturing industries, steel combines high TFP growth with massive downsizing and falling prices.
- This table motivates the paper's main puzzle: how can an industry shrink so much and yet become dramatically more productive?

TABLE 1—RELATIVE PERFORMANCE OF THE STEEL SECTOR (*Percent*)

	Steel sector	Mean sector	Median sector
Δ TFP	28	7	3
Δ shipments	−35	60	61
Δ labor	−80	−5	−1
Δ materials	−41	45	39
Δ value added	−43	34	38
Δ price†	−23	−2	−3
Δ material price†	−10	−11	−9

Notes: Only sectors over ten billion dollars are included. Changes computed between 1972–2002.
†Material and output prices indexes are deflated by the GDP deflator.
Source: NBER-CES Dataset for SIC Code 3312.

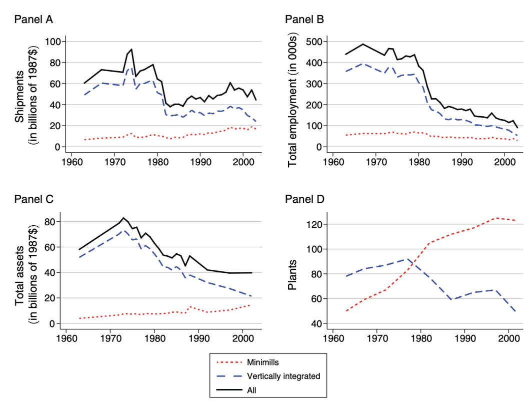

FIGURE 1. EVOLUTION OF THE STEEL INDUSTRY AND VERTICALLY INTEGRATED MILLS AND MINIMILLS

5.2 Figure 1: Evolution of the industry and the two technologies

Read:

- Total shipments recover somewhat after the early-1980s collapse, but employment keeps falling. This is the core descriptive fact behind measured productivity growth.

- The number of MM plants rises steadily while the number of VI plants falls, showing technology diffusion by extensive margin.
- Total assets also decline, which reinforces that the steel productivity story is not just simple capital deepening.

5.3 Figure 2: Minimill market share by major product

Read:

- Minimills take over low-quality products first — especially hot rolled bars and semifinished products — but remain minor players in sheet until much later.
- This product segmentation is essential for the competition mechanism: integrated plants face head-to-head minimill competition mainly in bars, not in sheet.
- The figure foreshadows the later result that surviving integrated plants tilt toward sheet products.

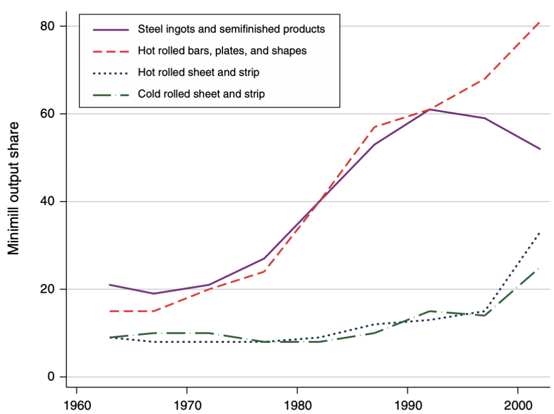

Figure 2. Minimills Market Share by Major Product

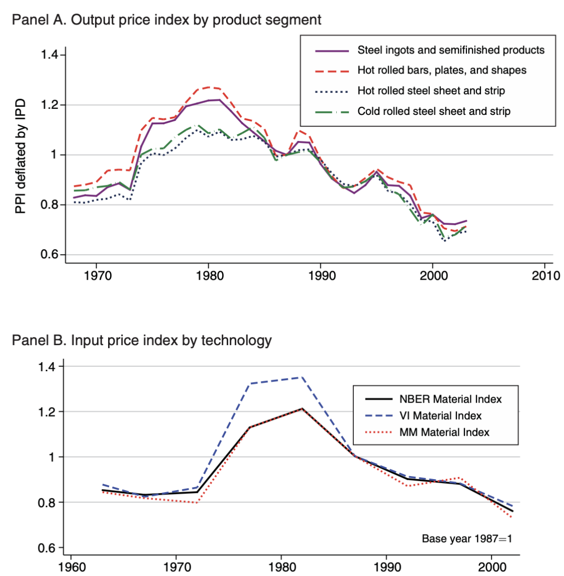

Figure 3. Producer Output and Input Price Indexes

Notes: Producer price index for selected steel products deflated by GDP deflator. Material price index deflated by GDP deflator. Base year 1987 = 100.

Source: BLS and authors' calculations

5.4 Figure 3 and Table 2: Price patterns and simultaneous entry-exit

Read:

- Output prices fall sharply after the early 1980s, and bar-product prices fall especially strongly. That is consistent with intensified competition where minimills expand most.
- Input prices diverge across technologies during the energy-price spike because VI plants are more energy intensive.

- Table 2 shows that in the 1980s the industry experiences exactly the kind of simultaneous minimill entry and integrated-plant exit that should matter for aggregate productivity.

Table 2—Entry and Exit in US Steel

	Entrant market share (plants)	Exiter market share (plants)
1963–1972	6 (29)	9 (D∗)
1973–1982	5 (49)	20 (20)
1983–1992	21 (55)	18 (47)
1993–2002	12 (30)	2 (41)
Minimills	Entrants	Exiters
1963–1972	17	D∗
1973–1982	39	0
1983–1992	43	26
1993–2002	D∗	17
Vertically integrated	Entrants	Exiters
1963–1972	12	D∗
1973–1982	10	20
1983–1992	12	21
1993–2002	D∗	24

Notes: D∗ cannot be disclosed due to the small number of observations. Numbers refer to the revenue market share represented by exiters and entrants, while numbers in parentheses refer to the count of plants that enter or exit.

Table 3—Production Function Coefficients and Technology Premium (*Percent*)

	Input and output price deflators		No plant-level output price deflator		No plant-level input price deflator	
	GMM (1)	OLS (2)	GMM (3)	OLS (4)	GMM (5)	OLS (6)
Material	0.680	0.631	0.650	0.610	0.680	0.631
	[0.65 0.73]	[0.58 0.69]	[0.62 0.70]	[0.52 0.67]	[0.64 0.73]	[0.58 0.69]
Labor	0.274	0.327	0.282	0.332	0.273	0.327
	[0.24 0.31]	[0.28 0.37]	[0.24 0.32]	[0.29 0.38]	[0.24 0.31]	[0.28 0.37]
Capital	0.079	0.050	0.082	0.051	0.082	0.050
	[0.04 0.11]	[−0.01 0.10]	[0.05 0.11]	[−0.01 0.10]	[0.05 0.11]	[−0.01 0.10]
VI premium	−0.075	−0.018	−0.038	0.013	−0.076	−0.018
	[−0.12 −0.04]	[−0.04 0.00]	[−0.08 0.00]	[−0.01 0.03]	[−0.12 −0.04]	[−0.04 0.00]

Notes: Ninety-five percent confidence in parentheses. GMM indicates that a two-stage investment control function procedure with a selection adjustment was used. VI premium is calculated by projecting estimated productivity on an indicator for a vertically integrated plant, along with year dummies. Standard errors are clustered at the plant-level to control for heteroskedasticity and serial correlation. For GMM columns, this clustering is computed via block bootstrap, which, in addition, corrects for the multi-step nature of the GMM estimator.

5.5 Table 3 and Table 4: Production-function estimates and the minimill premium

Read:

- The preferred GMM specification with plant-level price corrections implies a minimill productivity premium of about 7.5%.
- Output-price corrections are quantitatively important; input-price corrections matter less for the cross-technology productivity gap.
- Allowing technology-specific elasticities does not overturn the result, suggesting the main difference is in productivity levels and not sharply different factor elasticities.

Table 4—Production Functions: Minimills versus Vertically Integrated Plants

	OLS		GMM	
	Pooled (1)	Tech-specific (2)	Pooled (3)	Tech-specific (4)
Material	0.608***	0.614***	0.680***	0.657***
	(0.034)	(0.032)	(0.022)	(0.026)
Labor	0.335***	0.306***	0.274***	0.261***
	(0.028)	(0.023)	(0.020)	(0.022)
Capital	0.043	0.056	0.079***	0.097***
	(0.037)	(0.029)	(0.016)	(0.022)
Material × VI		−0.029		0.059
		(0.063)		(0.047)
Labor × VI		0.072		0.008
		(0.049)		(0.042)
Capital × VI		−0.031		−0.034
		(0.023)		(0.030)
Year FE	X	X		
R^2	0.966	0.966		
F-stat	1,663.267	1,426.796		
Plants	1,498	1,498	1,498	1,498

Note: VI is a dummy variable equal to one if a plant is a vertically integrated producer.

Table 5—Technology Premium: Robustness Checks

	Productivity ω			
	(1)	(2)	(3)	(4)
VI	−0.074**	−0.084	−0.102***	−0.178*
	(0.024)	(0.048)	(0.020)	(0.074)
Fixed effect				
Year	X			
Year-firm		X		
State-year			X	
Firm-year-state				X
Observations	1,498	1,498	1,498	1,498
Groups	301	914	289	1,291
R^2	0.052	0.005	0.021	0.027
F-stat	15.220	3.006	25.373	5.730

Notes: The standard errors are clustered at the plant-level. VI indicates vertically integrated plants.

5.6 Table 5 and Table 6: Management, geography, trade, and unions

Read:

- The minimill premium survives firm-year, state-year, and firm-state-year fixed effects, which weakens management and geography explanations.
- Steel does not experience unusually large import competition relative to manufacturing, so trade is not the main explanation for the sector's exceptional productivity growth.
- The unionization discussion points in the same direction: the technology effect is too robust to be mainly a union story.

TABLE 6—INTERNATIONAL COMPETITION: COMPARING THE STEEL SECTOR TO US MANUFACTURING

	Change total imports		
Period	Steel	Average	Median
Percent Δ [72–02]	4.1	66.1	23.7
	Import penetration rate		
Year	Steel	Average	Weighted average
1972	0.099	0.066	0.055
1996	0.180	0.220	0.157
Change	0.081	0.154	0.102
Percent change	81	233	85

Notes: The average and weighted average are computed over all available four-digit SIC87 industries using data provided by the NBER Manufacturing Database (for shipment data) and Bernard, Jensen, and Schott (2006) (for import penetration data). The changes are computed by industry before taking averages. The import penetration rate data are available only up to 1996. Weights are based on the industry's share of shipments in total manufacturing shipments as recorded in the NBER Manufacturing Database.

TABLE 7—STATIC DECOMPOSITIONS OF PRODUCTIVITY GROWTH CHANGE 1963–2002 (*Percent*)

Aggregate TFP $\Delta\Omega$	22.1	
Olley-Pakes decomposition:		
Unweighted average: $\Delta\,\overline{\omega}$	15.7 (0.71)	
Covariance: $\Delta\,\Gamma^{OP}$	6.4 (0.29)	
Between decomposition:		
Unweighted average: $\Delta\,\overline{\Omega}$	17.0 (0.77)	
Between covariance: $\Delta\,\Gamma^{B}$	5.1 (0.23)	
Within decomposition:	Minimills	Integrated
Aggregate TFP: $\Delta\,\Omega(\psi)$	9.6	24.3
Unweighted average: $\Delta\,\overline{\omega}(\psi)$	5.4 (0.55)	18.4 (0.83)
Within covariance: $\Delta\,\Gamma^{OP}(\psi)$	4.4 (0.45)	3.7 (0.17)

Note: The share of each component in the total aggregate productivity growth is listed in parentheses.

5.7 Table 7: Static decomposition of productivity growth

Read:

- Aggregate TFP growth is about 22.1% from 1963 to 2002 in the plant-level estimates.
- The Olley-Pakes covariance term contributes about 6.4 percentage points, meaning that reallocation toward more productive plants is economically important.
- The between-technology covariance contributes about 5.1 percentage points — around 23% of total productivity growth — showing that reallocation from VI to MM matters directly.

5.8 Table 8: Dynamic decomposition

Read:

- For the whole industry, plant improvement contributes about 9.5 points, reallocation about 9.3, and net entry about 3.3.
- Minimill productivity growth comes almost entirely from plant improvement, consistent with learning and technological progress within the new technology.
- Integrated-plant productivity growth comes heavily from reallocation and net entry/exit, consistent with competition-induced selection among incumbents.

5.9 Table 9 and Table 10: Why integrated plants catch up

Read:

Table 8—Dynamic Decomposition of Productivity Growth (*Percent*)

Component	All	Minimill	Integrated
Total change	22.1%	9.6	24.3
		(0.28)	(0.49)
Plant improvement	9.5%	11.8	9.3
		(0.34)	(0.19)
Reallocation	9.3%	−0.3	11.3
		(−0.03)	(0.23)
Net entry	3.3%	−2.0	3.8
		(−0.03)	(0.07)
Entry-exit premium		0.0	4.4

Notes: The share of each component in the total aggregate productivity growth is listed in parentheses. See equation (17) for definitions of various terms. For example, the share of minimill productivity growth (9.6 percent) in aggregate productivity growth is given by: $9.6/17.7 \times 0.77 = 0.28$—i.e., we compute the share of the minimill productivity growth in the unweighted aggregate productivity growth term, which we know from the top panel is 0.77.

Table 9—Sheet Productivity Premium

	Productivity ω							
	Panel A. All plants				*Panel B. Vertically integrated*			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Vertically integrated	−0.080***	−0.082***	−0.115***	−0.089*				
	(0.021)	(0.024)	(0.018)	(0.036)				
Sheet specialization ratio		0.073*	0.094***	0.119**	0.058	0.099**	0.114*	0.168*
		(0.035)	(0.025)	(0.036)	(0.038)	(0.032)	(0.047)	(0.084)
Fixed effects								
Year	X	X			X			X
State-year			X			X		
Firm-year				X			X	
Plant								X
Observations	1,498	1,498	1,498	1,498	667	667	667	667
Number of FE	301	301	289	914	124	175	353	124
R^2	0.020	0.082	0.034	0.024	0.069	0.019	0.019	0.081
F-stat	14.639	17.167	21.143	7.019	7.944	9.290	5.980	5.214

Notes: Standard errors clustered by plant, but not corrected for the sampling error in constructed productivity. Sheet specialization ratio is the share of revenues accounted for by sheet products (hot and cold rolled sheet).

- Sheet-specialized plants are more productive than bar-specialized plants, including within the integrated technology.
- Plants more specialized in sheet are also much more likely to survive. A fully sheet-specialized plant has roughly a 31 percentage point higher survival probability than a fully bar-specialized plant.
- This is the core evidence behind the paper's "catch-up" story: integrated plants become more productive mainly because low-productivity bar producers exit.

Table 10—Determinants of Exit

	Plant exists in 2002				
	Panel A. All plants		*Panel B. Vertically integrated*		
	(1)	(2)	(3)	(4)	(5)
Vertically integrated	−0.36***	−0.39***			
	(0.09)	(0.08)			
Sheet specialization ratio	0.39**	0.36*	0.31*	0.31*	0.22
	(0.14)	(0.14)	(0.13)	(0.13)	(0.14)
log capital (k)		0.02		0.20	0.24
		(0.03)		(0.24)	(0.25)
Productivity (ω)		0.03			0.05
		(0.14)			(0.04)
Observations	128	128	78	78	78
log-likelihood	−73.88	−73.72	−40.36	−40.02	−39.24
χ^2	16.97	17.30	5.89	6.58	8.12
Baseline probability	0.33	0.33	0.23	0.23	0.22

Notes: Marginal effects presented. Dependent variable is whether the plant has not exited by 2002 given its status in 1963. Very similar results are found with 1972 and 1977 as base years.

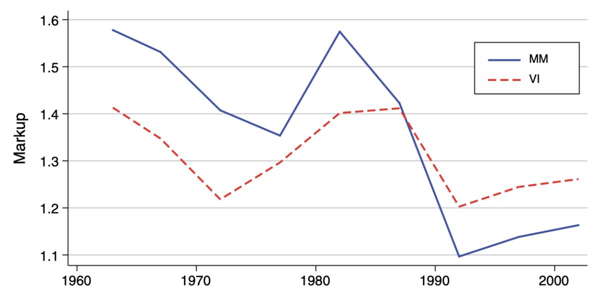

Figure 4. Market Share Weighted Markups

Source: Own calculations using US Census data.

5.10 Figure 4 and Tables 11–12: Markups and robustness

Read:

- Markups fall substantially over time: roughly from 1.8 to 1.1 for minimills and from 1.4 to about 1.2 for integrated plants.
- The larger markup decline for minimills is consistent with especially strong competition in the bar segment.
- The decomposition results are qualitatively robust across alternative productivity measures and also under the Petrin-Levinsohn alternative decomposition.

TABLE 11—STATIC DECOMPOSITIONS UNDER ALTERNATIVE PRODUCTIVITY MEASURES (*Percent*)

	Baseline (1)	OLS (2)	No material price deflator (3)	No output price deflator (4)	Tech-specific production function (5)
Change in TFP	22	28	22	23	19
	[0.17 0.27]	[0.23 0.33]	[0.16 0.27]	[0.18 0.28]	[0.01 0.27]
Olley-Pakes					
Unweighted average ($\Delta\overline{\omega}$)	71	69	71	77	65
	[0.59 0.79]	[0.62 0.73]	[0.60 0.79]	[0.68 0.84]	[−0.01 0.87]
Covariance ($\Delta\Gamma^{OP}$)	29	32	29	23	36
	[0.21 0.41]	[0.27 0.38]	[0.21 0.41]	[0.16 0.32]	[0.14 1.01]
Between					
Change in $\overline{\Omega}$	77	90	76	82	81
	[0.68 0.83]	[0.86 0.94]	[0.68 0.83]	[0.74 0.87]	[0.52 2.50]
Change in between technology	23	10	24	18	19
Covariance ($\Delta\Gamma^{B}$)	[0.17 0.32]	[0.06 0.14]	[0.17 0.33]	[0.13 0.26]	[−1.50 0.49]
Within technology					
Minimills					
Change in TFP	43	74	42	50	45
$\left(\text{ratio to aggregate } \frac{\Delta\Omega(MM)}{\Delta\Omega}\right)$	[0.19 0.60]	[0.59 0.85]	[0.18 0.58]	[0.29 0.63]	[0.02 1.85]
Fraction from unweighted average	55	62	55	77	55
($\Delta\overline{\omega}(MM)$)	[−0.02 0.70]	[0.53 0.69]	[−0.05 0.70]	[0.59 0.86]	[−0.05 0.77]
Fraction from covariance	45	38	45	24	45
($\Delta\Gamma^{OP}(MM)$)	[0.31 1.02]	[0.31 0.47]	[0.30 1.05]	[0.14 0.41]	[0.23 1.05]
Vertically integrated					
Change in TFP	110	107	111	114	118
$\left(\text{ratio to aggregate } \frac{\Delta\Omega(VI)}{\Delta\Omega}\right)$	[1.03 1.21]	[1.01 1.14]	[1.03 1.22]	[1.06 1.24]	[0.78 3.24]
Fraction from unweighted average	83	76	84	89	87
($\Delta\overline{\omega}(VI)$)	[0.73 0.94]	[0.71 0.82]	[0.74 0.94]	[0.81 0.98]	[0.73 1.02]
Fraction from covariance	17	24	17	11	13
($\Delta\Gamma^{OP}(VI)$)	[0.06 0.27]	[0.18 0.30]	[0.06 0.26]	[0.02 0.19]	[−0.02 0.27]

Notes: GMM refers to the Olley-Pakes control function approach. Plant-level output prices refers to deflating revenue using product-level price indexes. Plant-level material prices refers to deflating material inputs costs using material specific price indexes. Bootstrapped 95 percent confidence intervals using 10,000 replications are shown in brackets, and these include only sampling error in the computation of productivity ω.

TABLE 12—DYNAMIC DECOMPOSITIONS UNDER ALTERNATIVE PRODUCTIVITY MEASURES (*Percent*)

	Baseline (1)	OLS (2)	No material price deflator (3)	No output price deflator (4)	Tech-specific production function (5)
Dynamic decomposition					
All					
Plant improvement	43	46	43	49	47
	[0.35 0.49]	[0.40 0.50]	[0.34 0.49]	[0.43 0.53]	[0.20 1.15]
Reallocation	42	42	42	40	43
	[0.35 0.54]	[0.37 0.49]	[0.35 0.54]	[0.33 0.49]	[0.30 0.83]
Entry-exit	15	12	15	11	10
	[0.12 0.18]	[0.11 0.13]	[0.12 0.18]	[0.08 0.14]	−[0.85 0.29]
Minimills					
Plant improvement	123	87	125	136	124
	[0.94 2.59]	[0.77 1.03]	[0.94 2.68]	[1.07 2.36]	[0.88 2.59]
Reallocation	−3	5	−3	−11	9
	−[0.51 0.09]	−[0.03 0.12]	−[0.54 0.09]	−[0.49 0.01]	−[0.56 1.03]
Entry-exit	−21	8	−22	−25	−34
	−[1.08 0.00]	−[0.02 0.14]	−[1.18 −0.01]	−[0.87 −0.07]	−[1.92 0.04]
Vertically Integrated					
Plant improvement	38	40	38	42	39
	[0.26 0.48]	[0.33 0.46]	[0.26 0.48]	[0.32 0.51]	[0.09 0.49]
Reallocation	46	54	46	46	51
	[0.33 0.61]	[0.47 0.61]	[0.32 0.61]	[0.34 0.59]	[0.32 1.15]
Entry-exit	16	7	16	12	11
	[0.10 0.22]	[0.05 0.09]	[0.11 0.22]	[0.07 0.17]	−[0.35 0.29]

Notes: GMM refers to the Olley-Pakes control function approach. Plant-level output prices refers to deflating revenue using product-level price indexes. Plant-level material prices refers to deflating material inputs costs using material specific price indexes. Bootstrapped 95 percent confidence intervals using 10,000 replications are shown in brackets, and these include only sampling error in the computation of productivity ω.

6 Short summary

- **Conclusion.** The rise of minimills is the central force behind the steel industry's productivity growth.
- **Mechanism.** Minimills raise productivity directly because they are better technology and indirectly because they intensify competition, which reallocates output away from inefficient integrated plants.
- **Quantitative lesson.** Reallocation is not an abstract residual. In this industry it can be tied to a concrete technological shock and traced through product-market competition, entry, exit, and changing markups.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** A drastic new process technology, the minimill, explains a large share of steel-sector productivity growth.
- **Direct effect.** Minimills are more productive than integrated plants and gain market share over time.
- **Indirect effect.** Their expansion raises competition, lowers markups, and forces reallocation within the incumbent VI technology.
- **Magnitude.** The paper attributes about 23% of aggregate productivity growth to between-technology reallocation and about another fifth to within-minimill productivity growth. To-

gether, this is about half of total industry productivity growth.

- **Welfare implication.** Back-of-the-envelope calculations suggest minimill-driven price declines raised consumer surplus by about \$9–11 billion per year.

7.2 Economic intuition

- A better technology does not need to replace the old one instantly to transform an industry.
- The old technology can survive, but only after being disciplined by competition and stripped of its weakest producers.
- Because steel products are relatively stable, the link from process innovation to reallocation and aggregate productivity is unusually transparent in this setting.

7.3 Takeaway

This paper is a clean demonstration that technology and reallocation are tightly linked. The minimill matters not only because it is more efficient, but because its arrival changes who survives, who expands, which product segments become competitive, and how much market power firms can sustain. The steel industry therefore provides unusually clear evidence that process innovation can reshape aggregate productivity through both direct plant-level improvements and equilibrium reallocation across producers.